

Section 7.1: Introduction

Def. A statistic is a function of

1. random variables (e.g., capital letters from end of alphabet)
- &
2. known constants

Some examples:

1. Sample mean: $\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i$ ↗ sample size

2. Sample variance: $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$

3. Order statistics

→ $Y_{(1)} = \min(Y_1, Y_2, \dots, Y_n)$

→ $Y_{(n)} = \max(Y_1, Y_2, \dots, Y_n)$

Def. The probability distribution of a statistic is called a sampling distribution.

Example 1. A fair six-sided die is rolled three times. Let Y_1 , Y_2 , and Y_3 denote the outcome of rolls 1, 2, and 3, respectively.

(a) Find the probability distribution, mean, and variance for one roll.

Let $Y_i = \text{outcome for die } i$

probability! ↗

Y_i	1	2	3	4	5	6
$P(Y_i)$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$	$1/6$

↖ $\mu_{Y_i} = E(Y_i) = \sum_{i=1}^6 Y_i \cdot P(Y_i) = 1(1/6) + 2(1/6) + \dots + 6(1/6) = \underline{\underline{3.5}}$

Greek letter "mu"

$$\sigma_{Y_i}^2 = V(Y_i) = E(Y_i^2) - [E(Y_i)]^2 = E[(Y_i - \mu)^2] = \sum_{i=1}^6 (Y_i - \mu)^2 P(Y_i) = (1-3.5)^2(1/6) + \dots + (6-3.5)^2(1/6) = \underline{\underline{2.9167}}$$

Hardwork 21
(MAT 311)

(b) Find the mean of $\bar{Y}$, $\mu_{\bar{Y}}$.

$$\mu_{\bar{Y}} = E(\bar{Y}) = E\left(\frac{1}{n} \sum_{i=1}^n Y_i\right) = \frac{1}{n} \sum_{i=1}^n E(Y_i) = \frac{1}{n} \cdot n \cdot 3.5 = 3.5$$

→ Ex. 1: $\mu_{\bar{Y}} = 3.5$ as well!

(c) Find the standard deviation of $\bar{Y}$, $\sigma_{\bar{Y}}$.

$$\sigma_{\bar{Y}} = SD(\bar{Y}) = \sqrt{V\left(\frac{1}{n} \sum_{i=1}^n Y_i\right)} \stackrel{\text{independence}}{=} \sqrt{\frac{1}{n^2} \sum_{i=1}^n V(Y_i)} = \sqrt{\frac{1}{n^2} \cdot n \cdot 2.9167} = \frac{1.7078}{\sqrt{n}}$$

→ Ex. 1: $\sigma_{\bar{Y}} = \frac{1.7078}{\sqrt{3}} \approx \boxed{.986}$

(d) How could we find the sampling distribution of $\bar{Y}$?

Let $W = Y_1 + Y_2 + Y_3$. Then W can take the values $3, 4, \dots, 18$.

$$\bar{Y} = \frac{W}{3}.$$

$$P(W=3) = P\left(\bar{Y} = \frac{3}{3} = 1\right) = P((1, 1, 1)) = \left(\frac{1}{6}\right)^3 = \frac{1}{216}$$

$$P(W=4) = P\left(\bar{Y} = \frac{4}{3}\right) \stackrel{\text{can add since we have disjoint events}}{=} P((1, 1, 2)) + P((1, 2, 1)) + P((2, 1, 1)) = \frac{3}{216}$$

⋮

$$P(W=18) = \frac{1}{216}$$

Question. How can we get an idea about the shape of the sampling distribution of $\bar{Y}$ without doing all of these calculations?

Simulate it! Take repeated independent samples of size 3, then calculate $\bar{y}$ for each one, and construct a histogram of all the $\bar{y}$ values.

Section 7.2: Sampling Distributions Related to the Normal Distribution

Important Abbreviation. *i.i.d.* means independent and identically distributed

Theorem 7.1. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, $E(\bar{Y})$ $SD(\bar{Y})$

$$\bar{Y} = \frac{1}{n} \sum_{i=1}^n Y_i \sim N\left(\mu, \frac{\sigma}{\sqrt{n}}\right).$$

Proof. Recall the following theorem from Handout 25 (MAT 310): If $Y_1, Y_2, \dots, Y_n$ are independent Normal RVs such that $Y_i \sim N(\mu_i, \sigma_i)$ and $X = \sum_{i=1}^n \overbrace{a_i}^{\text{constants}} Y_i$, then

$$X \sim N\left(\sum_{i=1}^n a_i \mu_i, \sqrt{\sum_{i=1}^n a_i^2 \sigma_i^2}\right).$$

Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$; i.e., $\mu_i = \mu$ and $\sigma_i = \sigma$ for all i .

Define $a_i = \frac{1}{n}$ for all i .

$$\text{Then, } X = \sum_{i=1}^n \left(\frac{1}{n}\right) Y_i = \frac{1}{n} \sum_{i=1}^n Y_i = \bar{Y}.$$

Thus, applying the above theorem, $\bar{Y} \sim N\left(\sum_{i=1}^n \left(\frac{1}{n}\right) \mu, \sqrt{\sum_{i=1}^n \left(\frac{1}{n}\right)^2 \sigma^2}\right)$

$$\Rightarrow \bar{Y} \sim N\left(\mu, \sigma/\sqrt{n}\right) \quad \square$$

Fact 1. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right)$. Then,

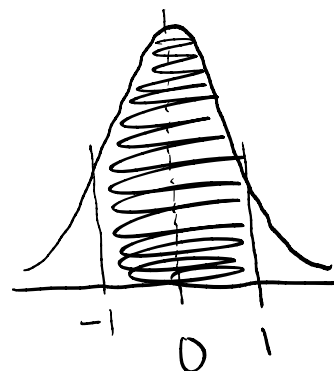
$$Z \sim N(0, 1).$$

Example 2. Suppose that light bulbs have a length of life that is normally distributed with mean μ and standard deviation $\sigma = 40$ hours. Suppose we take a random sample of 16 light bulbs and measure their lengths of life.

$n \nearrow$

(a) Find the probability that the sample mean will be within 10 hours of the population mean.

$$\begin{aligned} P(|\bar{Y} - \mu| \leq 10) &= P(-10 \leq \bar{Y} - \mu \leq 10) \\ &= P\left(\frac{-10}{40/\sqrt{16}} \leq \frac{\bar{Y} - \mu}{40/\sqrt{16}} \leq \frac{10}{40/\sqrt{16}}\right) \\ &= P(-1 \leq Z \leq 1) \\ &= \text{normalcdf}(-1, 1, 0, 1) \\ &\approx \boxed{0.6827} \end{aligned}$$

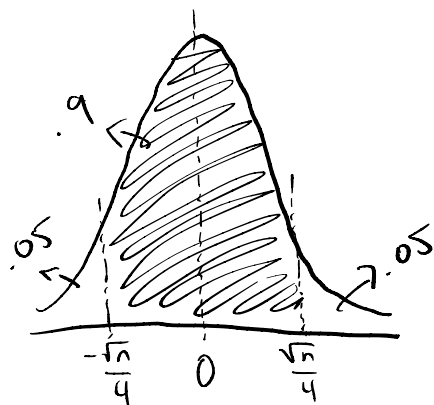


lower bound upper bound mean std. deviation

(b) Suppose we would like the sample mean to be within 10 hours of the population mean with probability 0.90. How many light bulbs must we sample in order to ensure this degree of accuracy?

WTF! $\odot$ s.t. $P(|\bar{Y} - \mu| \leq 10) = .9$.

$$\begin{aligned} P(|\bar{Y} - \mu| \leq 10) &= P(-10 \leq \bar{Y} - \mu \leq 10) \\ &= P\left(\frac{-10}{40/\sqrt{n}} \leq Z \leq \frac{10}{40/\sqrt{n}}\right) \\ &= P\left(-\frac{1}{4}\sqrt{n} \leq Z \leq \frac{1}{4}\sqrt{n}\right) = .9 \end{aligned}$$



$$\rightarrow \frac{\sqrt{n}}{4} = \text{invNorm}(.95, 0, 1) \approx 1.645$$

$$\rightarrow \sqrt{n} = 6.58$$

$$\rightarrow n = (6.58)^2 \approx 43.2964$$

$$\rightarrow \text{Use } \boxed{n \geq 44}$$

Fact 2. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z_i = \frac{Y_i - \mu}{\sigma}$. Then,

$$Z_i^2 \sim \chi_1^2 \quad (\text{chi-square distribution with 1 degree of freedom})$$

Theorem 7.2. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $Z_i = \frac{Y_i - \mu}{\sigma}$. Then,

$$\sum_{i=1}^n Z_i^2 \sim \chi_n^2$$

Proof.

$$\text{Let } W = \sum_{i=1}^n Z_i^2.$$

$$\begin{aligned} \text{Using mgfs, } m_W(t) &= m_{Z_1^2}(t) \cdot m_{Z_2^2}(t) \cdot \dots \cdot m_{Z_n^2}(t) \quad \rightarrow \text{Handout 25 from MAT 310} \\ &= (1-2t)^{-1/2} \cdot (1-2t)^{-1/2} \cdot \dots \cdot (1-2t)^{-1/2} \\ &= \underbrace{(1-2t)^{-n/2}}_{\text{mgf of a } \chi_n^2 \text{ RV}} \end{aligned}$$

Thus, $W \sim \chi_n^2$ by uniqueness of mgfs. $\square$

Question. How can we use software to find probabilities for $Y \sim \chi_\nu^2$?

- $P(Y \leq y) = \chi^2 \text{cdf}(0, y, \nu)$

- $P(Y > y) = 1 - P(Y \leq y) = 1 - \chi^2 \text{cdf}(0, y, \nu)$

Question. How can we find the $(100 \cdot p)^{th}$ percentile, ϕ_p , of $Y \sim \chi^2_\nu$? Table 6 in Appendix 3!

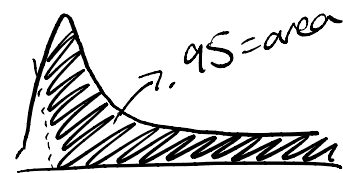
• $\phi_{0.95}$ of a χ^2_6 RV: 12.5916

• $\phi_{0.10}$ of a χ^2_{21} RV: 13.2396

Example 3. Suppose $Z_1, Z_2, \dots, Z_{10} \stackrel{i.i.d.}{\sim} N(0, 1)$.

(a) Find b such that $P\left(\sum_{i=1}^{10} Z_i^2 \geq b\right) = 0.95$.

Note: $\sum_{i=1}^{10} Z_i^2 \sim \chi^2_{10}$ (Thm. 7.2)



$\chi^2_{.95}$ with 10 df = 3.9403

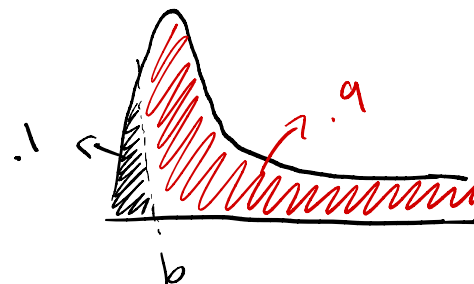
(b) Find b such that $P\left(\sum_{i=1}^{10} Z_i^2 < b\right) = 0.10$.

$b = 4.86518$

↑

$\chi^2_{.9}$ with 10 df

(Table 6)



pg. 358 of textbook

Fact 3. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then, $\bar{Y}$ and $S^2 = \frac{1}{n-1} \sum_{i=1}^n (Y_i - \bar{Y})^2$ are independent.

Theorem 7.3. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then,

$$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$$

Proof.

Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$. Then $\bar{Y} \sim N(\mu, \sigma/\sqrt{n})$.
 Let $Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right) \sim N(0, 1)$.

Thus, $Z^2 = n \left(\frac{\bar{Y} - \mu}{\sigma} \right)^2 \sim \chi^2_1$ (by a previous Fact).

Define $W = \sum_{i=1}^n Z_i^2$, where $Z_i = \frac{Y_i - \mu}{\sigma}$. Then, $W \sim \chi^2_n$.
Thm. 7.2

We can rewrite W as $W = \underbrace{\sum_{i=1}^n \left(\frac{Y_i - \bar{Y}}{\sigma} \right)^2}_{\text{EXERCISE}} + n \left(\frac{\bar{Y} - \mu}{\sigma} \right)^2$
 $= \frac{(n-1)S^2}{\sigma^2} + Z^2$.

From Fact 3, $\bar{Y}$ and S^2 are independent, so $\bar{Y}$ and $\frac{(n-1)S^2}{\sigma^2}$ are independent. Thus, Z^2 and

$\frac{(n-1)S^2}{\sigma^2}$ are independent (since the only random

part of Z^2 is $\bar{Y}$). Using mgfs, we know

$$m_W(t) = m_{\frac{(n-1)S^2}{\sigma^2}}(t) \cdot m_{Z^2}(t) \cdot (1-2t)^{-\frac{(n-1)}{2}}$$

$$\rightarrow (1-2t)^{-n/2} = m_{\frac{(n-1)S^2}{\sigma^2}}(t) \cdot (1-2t)^{-1/2} \rightarrow m_{\frac{(n-1)S^2}{\sigma^2}}(t) = \leftarrow$$

$$\sum_{i=1}^n \left(\frac{y_i - \bar{y}}{\sigma} \right)^2 + n \left(\frac{\bar{y} - \mu}{\sigma} \right)^2$$

$$= \frac{1}{\sigma^2} \left[\sum_{i=1}^n (y_i^2 - 2y_i\bar{y} + \bar{y}^2) + n(\bar{y}^2 - 2\mu\bar{y} + \mu^2) \right]$$

$$= \frac{1}{\sigma^2} \left[\sum_{i=1}^n y_i^2 - \underbrace{2\bar{y} \sum_{i=1}^n y_i}_{2\bar{y} \cdot n \cdot \frac{1}{n} \sum y_i = 2\bar{y} n \bar{y} = 2n\bar{y}^2} + \cancel{n\bar{y}^2} + \cancel{n\bar{y}^2} - 2n\mu\bar{y} + n\mu^2 \right]$$

$$2\bar{y} \cdot n \cdot \frac{1}{n} \sum y_i = 2\bar{y} n \bar{y} = 2n\bar{y}^2$$

$$= \frac{1}{\sigma^2} \left[\sum_{i=1}^n y_i^2 - 2n\mu\bar{y} + n\mu^2 \right]$$

$$= \sum_{i=1}^n \left(\frac{y_i - \mu}{\sigma} \right)^2$$

$$= \sum_{i=1}^n z_i^2$$

$$2\mu \sum_{i=1}^n y_i$$

Example 2 cont'd. Suppose that light bulbs have a length of life that is normally distributed with mean μ and standard deviation $\sigma = 40$ hours. Suppose we take a random sample of 16 light bulbs and measure their lengths of life.

(c) Find an interval that will contain S^2 with probability 0.95.

$$Y \sim N(\mu, \sigma = 40)$$

WTF: an interval (b_1, b_2) such that $P(b_1 \leq S^2 \leq b_2) = .95$.

$$P(b_1 \leq S^2 \leq b_2) = .95 \rightarrow P\left(\frac{(n-1)}{\sigma^2} b_1 \leq \frac{(n-1)S^2}{\sigma^2} \leq \frac{(n-1)}{\sigma^2} b_2\right) = .95$$

$\sim \chi_{n-1}^2 = \chi_{15}^2$

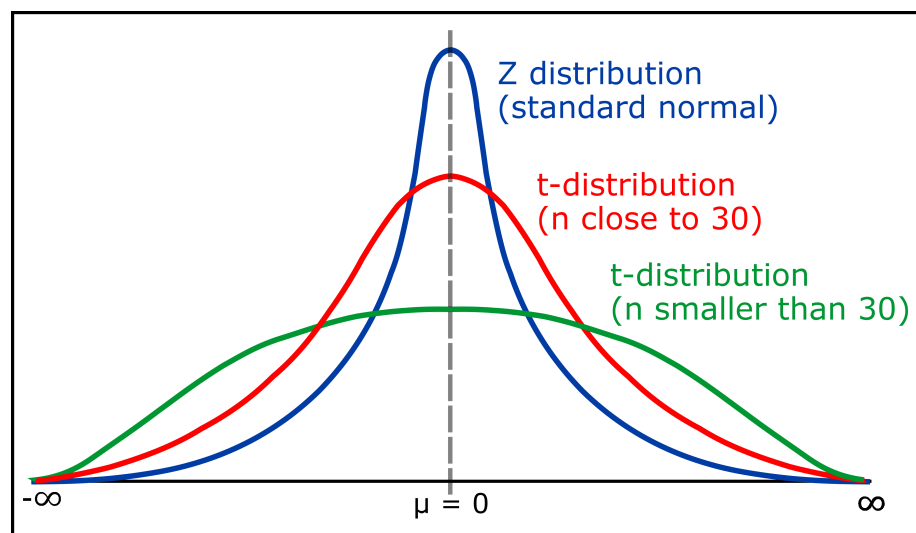
$$\frac{(n-1)b_1}{\sigma^2} = \frac{15(b_1)}{1600} = 6.26214 \rightarrow b_1 = 667.9616$$

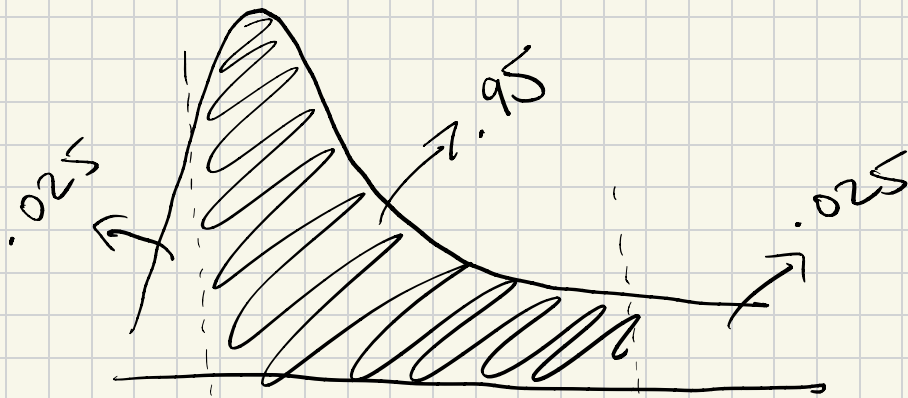
$$\frac{(n-1)b_2}{\sigma^2} = \frac{15b_2}{1600} = 27.4884 \rightarrow b_2 = 2932.096$$

Remark. Theorem 7.1 provides the basis for inference-making procedures about μ for a Normal population with standard deviation, σ .

Question. What if σ is unknown?

→ We can use S to estimate σ & the t -distribution.





$$\frac{(n-1)b_1}{\sigma^2}$$

↑

$$= \chi^2_{.975} = 6.26214$$

$$\frac{(n-1)b_2}{\sigma^2}$$

↑

$$= \chi^2_{.025} = 27.4884$$

Fact 4. Suppose $Z \sim N(0, 1)$ and $W \sim \chi_\nu^2$. Then, if Z and W are independent,

$$T = \frac{Z}{\sqrt{W/\nu}} \sim t_\nu \quad \text{degrees of freedom}$$

Fact 5. Let $Y_1, Y_2, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$ and $T = \frac{\bar{Y} - \mu}{S/\sqrt{n}} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right)$. Then,

$$T \sim t_{n-1}.$$

Proof. Suppose $Y_1, \dots, Y_n \stackrel{i.i.d.}{\sim} N(\mu, \sigma)$.

Define $Z = \sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right) \sim N(0, 1)$.

Define $W = \frac{(n-1)S^2}{\sigma^2} \sim \chi_{n-1}^2$.

Note that Z and W are independent.

$$T = \frac{Z}{\sqrt{W/(n-1)}} = \frac{\sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right)}{\sqrt{\frac{(n-1)S^2}{\sigma^2(n-1)}}} = \frac{\sqrt{n} \left(\frac{\bar{Y} - \mu}{\sigma} \right) \cdot \frac{\sigma}{S}}{1} = \sqrt{n} \left(\frac{\bar{Y} - \mu}{S} \right) \sim t_{n-1}. \quad \text{Fact 4}$$

Question. How can we use software to find probabilities for $Y \sim t_{n-1}$?

- $P(Y \leq y) = \text{tcdf}(-1E99, y, n-1)$

Question. How can we find the $(100 \cdot p)^{\text{th}}$ percentile, ϕ_p , of $Y \sim t_{n-1}$?

- $\phi_{0.95}$ of a t_6 RV: $\text{invT}(.95, 6) \approx 1.9432$
- $\phi_{0.10}$ of a t_{21} RV: $\text{invT}(.1, 21) \approx -1.3232$

Example 4. Suppose it is known that the GPAs of freshmen are normally distributed with unknown mean and variance, μ and σ^2 , respectively. We take a random sample of 8 freshmen. Find the probability that $\bar{Y}$ will be within $3S/\sqrt{8}$ of the population mean, μ .

Let $Y = \text{GPA of freshman}$. $Y \sim N(\mu, \sigma)$.

$$P\left(|\bar{Y} - \mu| \leq \frac{3S}{\sqrt{8}}\right) = P\left(\frac{-\frac{3S}{\sqrt{8}}}{\frac{S/\sqrt{n}}{\sqrt{n}}} \leq \frac{\bar{Y} - \mu}{\frac{S/\sqrt{n}}{\sqrt{n}}} \leq \frac{\frac{3S}{\sqrt{8}}}{\frac{S/\sqrt{n}}{\sqrt{n}}}\right)$$

$$\stackrel{n=8}{\rightarrow} P(-3 \leq T \leq 3)$$

$$= \text{tcdf}(-3, 3, 7)$$

$$\approx \boxed{.9801}$$

Fact 6. Let $W_1 \sim \chi_{\nu_1}^2$ and $W_2 \sim \chi_{\nu_2}^2$ be independent RVs. Then,

$$F = \frac{W_1/\nu_1}{W_2/\nu_2} \sim F_{\nu_1, \nu_2}$$

numerator df
denominator df

Question. Consider two independent random samples from normal distributions. One sample is size n_1 with variance σ_1^2 ; the other is size n_2 with variance σ_2^2 . Let S_1^2 and S_2^2 represent the estimators of σ_1^2 and σ_2^2 , respectively. What does the above definition imply about the distribution of $\frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2}$?

Define $W_i = \frac{(n_i-1)S_i^2}{\sigma_i^2} \sim \chi_{n_i-1}^2$, $i=1,2$. W_1 and W_2 are also independent!

$$F = \frac{W_1/(n_1-1)}{W_2/(n_2-1)} = \frac{[(n_1-1)S_1^2/\sigma_1^2]/(n_1-1)}{[(n_2-1)S_2^2/\sigma_2^2]/(n_2-1)} = \frac{S_1^2/\sigma_1^2}{S_2^2/\sigma_2^2} \sim F_{n_1-1, n_2-1}.$$

Question. How can we use software to find probabilities for $Y \sim F_{\nu_1, \nu_2}$?

- $P(Y \leq y) = \text{Fcdf}(0, y, \nu_1, \nu_2)$

Question. How can we find the $(100 \cdot p)^{\text{th}}$ percentile, ϕ_p , of $Y \sim F_{\nu_1, \nu_2}$?

- $\phi_{0.95}$ of a $F_{4,7}$ RV: 4.12
- $\phi_{0.995}$ of a $F_{2,9}$ RV: 10.11

Example 5. Suppose we have independent random samples from two normal populations with $n_1 = 16$, $\sigma_1^2 = 10$, $n_2 = 25$, and $\sigma_2^2 = 15$.

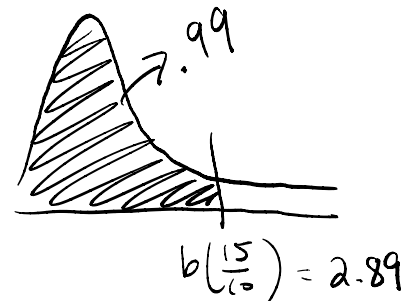
(a) Find the number b such that $P\left(\frac{S_1^2}{S_2^2} \leq b\right) = 0.99$.

$$P\left(\frac{S_1^2}{S_2^2} \leq b\right) = .99 \rightarrow P\left(\frac{S_1^2}{S_2^2} \cdot \frac{\sigma_2^2}{\sigma_1^2} \leq b \left(\frac{\sigma_2^2}{\sigma_1^2}\right)\right) = .99$$

$$\rightarrow P\left(F \leq b\left(\frac{15}{10}\right)\right) = .99$$

$$\rightarrow b\left(\frac{15}{10}\right) = 2.89$$

$$\rightarrow \boxed{b \approx 1.9267}$$



(b) Find the number c such that $P\left(\frac{S_1^2}{S_2^2} \geq c\right) = 0.99$.

$$\begin{aligned}
 P\left(\frac{S_1^2}{S_2^2} \geq c\right) &= .99 \rightarrow P\left(\frac{S_2^2}{S_1^2} \leq \frac{1}{c}\right) = .99 \\
 &\rightarrow P\left(\frac{S_2^2}{S_1^2} \cdot \frac{\sigma_1^2}{\sigma_2^2} \leq \frac{1}{c} \cdot \frac{\sigma_1^2}{\sigma_2^2}\right) = .99 \\
 &\rightarrow P\left(F \leq \frac{10}{15c}\right) = .99 \\
 &\rightarrow 3.29 = \frac{10}{15c} \\
 &\rightarrow \boxed{c \approx .2026}
 \end{aligned}$$

Section 7.3: The Central Limit Theorem

Question. What is the distribution of $\bar{Y}$ for i.i.d. Y_i that are not normal? Let's look at the following applet: https://digitalfirst.bfwpub.com/stats_applet/stats_applet_3_cltmean.html

Theorem 7.4: Central Limit Theorem (CLT). Let $Y_1, Y_2, \dots, Y_n$ be i.i.d. RVs with $E(Y_i) = \mu$ and $V(Y_i) = \sigma^2 < \infty$. Then, as $n \rightarrow \infty$,

$$Z = \frac{\bar{Y} - \mu}{\sigma/\sqrt{n}} \text{ converges in distribution to } \underline{N(0,1)}.$$

In other words, $\bar{Y}$ is asymptotically normal with mean μ and variance σ^2/n .

Remark. Regardless of the distribution of the i.i.d. Y_i , the CLT implies that $\bar{Y}$ is approximately normal for large n . But how large is "large enough"?

$$n \geq 30$$

Example 6. The weights of adult female elk are i.i.d. with a mean of 490 pounds and a standard deviation of 45 pounds. Compute the probability that the average weight of 100 female elk is above 500 pounds.

$\bar{Y}$ = avg. weight of female elk.

$$P(\bar{Y} > 500) \approx \text{normalcdf}(500, 1E99, 490, 45/\sqrt{100}) \\ \approx .0131$$

Remark. Review the CLT examples from MAT 310!

Other Important Distribution Results

Fact 7. Suppose $Y_1, \dots, Y_n \stackrel{\text{indep.}}{\sim} \text{Gamma}(\alpha_i, \beta)$. Let $U = \sum_{i=1}^n Y_i$. Then,

$$U \sim \text{Gamma}\left(\sum_{i=1}^n \alpha_i, \beta\right)$$

Proof.

$$m_U(t) = m_{Y_1}(t) \cdot m_{Y_2}(t) \cdot \dots \cdot m_{Y_n}(t) \quad [\text{by independence}]$$

$$= (1 - \beta t)^{-\alpha_1} \cdot (1 - \beta t)^{-\alpha_2} \cdot \dots \cdot (1 - \beta t)^{-\alpha_n}$$

$$= (1 - \beta t)^{-\sum_{i=1}^n \alpha_i}$$

$$\underbrace{\hspace{1cm}}_{\text{mgf of } \text{Gamma}(\sum_{i=1}^n \alpha_i, \beta)}$$

By uniqueness of mgfs, $U \sim \text{Gamma}(\sum_{i=1}^n \alpha_i, \beta)$. $\square$

Fact 8. Suppose $Y \sim \text{Gamma}(\alpha = \frac{n}{2}, \beta)$, where n is a positive integer. Let $W = \frac{2Y}{\beta}$. Then,

$$W \sim \chi_n^2$$